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Geochemical Evolution of Coastal Groundwater in Eastern Saudi Arabia: Insights from Lab-Scale Experiment and Inverse Geochemical Modeling

Bassam Tawabini, Geosciences Department, College of Petroleum Engineering & Geosciences, King Fahd University of Petroleum & Minerals, Dhahran, Saudi Arabia / Interdisciplinary Research Center for Membranes and Water Security, King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia; Mohammed Benaafi, Interdisciplinary Research Center for Membranes and Water Security, King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia / Geosciences Department, College of Petroleum Engineering & Geosciences, King Fahd University of Petroleum & Minerals, Dhahran, Saudi Arabia; Abdullah Basaleh and Abdullateef Ibrahim, Geosciences Department, College of Petroleum Engineering & Geosciences, King Fahd University of Petroleum & Minerals, Dhahran, Saudi Arabia

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Abstract

The study aims to understand the geochemical evolution of coastal groundwater in the eastern coastal region of Saudi Arabia using integrated lab-scale experimental and inverse geochemical modeling. The primary focus is on the geochemical processes occurring during seawater-groundwater mixing with emphasis on mineral dissolution/precipitation, cation exchange, adsorption, and trace element mobilization. 3D lab-scale physical experiments were carried out to simulate seawater intrusion in coastal aquifers using local aquifer materials, groundwater, and seawater. The lab experiments were conducted under various hydrogeological conditions and variable control factors to simulate groundwater dynamics in sandy coastal aquifers impacted by seawater intrusion. Post-experiment advanced geochemical analysis was performed to describe the chemical change of groundwater characteristics during seawater intrusion. Inverse geochemical modeling was performed to identify the geochemical processes that controlled the water chemistry during the seawater intrusion experiment. The lab experiment data were used to validate geochemical models. The integrated lab-scale experiment and inverse geochemical modeling reveal that the seawater intrusion in coastal aquifers has significantly impacted the groundwater quality through increasing salinity and trace element mobilization. The study showed that significant changes in groundwater quality occurred during low pumping and no recharge scenarios within high seawater mixing zones. The validated geochemical model offers a reliable prediction tool for groundwater quality under different hydrogeological scenarios. The study results underscore the importance of integrated lab experiments and modeling approaches to improve the understanding of seawater intrusion's impact on the geochemical evolution of coastal aquifers in arid regions.

Introduction

Due to increased human activity during the last century, coastal groundwater resources throughout the world have faced unprecedented water security challenges (du Plessis, 2023). Significant groundwater withdrawal in these regions has led to profound consequences on the groundwater ecosystem and several environmental issues, including groundwater contamination (Niazi et al., 2024). Human activities have significant control over the development of the groundwater ecosystem (M. Liu et al., 2024). Nevertheless, the consequences of various human activities and interactions between water and land are uncertain and complicated (Benaafi et al., 2023a, 2023b; Han et al., 2015). Investigating changes in groundwater quality induced by anthropogenic activities may aid in groundwater management in coastal locations. It is also essential for the long-term growth of water resources and the conservation of groundwater habitats. Natural mechanisms such as water-rock interactions and ion exchange have a significant role in determining the hydrogeochemical evolution of groundwater in coastal areas before human activities have an influence (Lambrakis, 2006; Mastrocicco & Colombani, 2021). Seawater intrusion (SWI) is the subsurface migration of seawater into freshwater aquifers, and it is a major environmental issue that poses a significant water security concern to coastal ecosystems. It occurs when seawater infiltrates landward as a result of substantial groundwater removal and/or rising sea levels (Setiawan et al., 2024). The global issue of SWI has attracted attention in the realm of groundwater management in coastal areas (Chang et al., 2024).

Moreover, urban, agricultural, and industrial development have a significant impact on the chemical composition of coastal groundwater (Benaafi et al., 2022; Li et al., 2021; Z. Wang et al., 2023). Quantifying the variations in hydrochemical domains of groundwater using numerical simulations has been implemented by many researchers (Benaafi et al., 2024; Cao et al., 2021; Gopinath et al., 2019; Zhang et al., 2015). In addition, several studies have documented the investigation of seawater intrusions (SWI) using numerical modeling and laboratory-scale experiments (Huang & Chiu, 2018; Morgan et al., 2013; Yu & Michael, 2019; Zhao et al., 2016). Nevertheless, these studies specifically investigated the dynamics of seawater intrusion and seawater-groundwater mixing zones. Several experimental and modeling studies have tested methods to control seawater intrusion in coastal aquifers. For example, Wang et al. (2022) combined sandbox tests with FEFLOW simulations and showed that adjusting the number and placement of recharge and pumping wells can push the saline wedge back from 20 cm² to 2 050 cm², raising aquifer recovery from 18.9% to 48.9%. Abarca and Clement (2009) developed a two-dimensional colorimetric tank paired with SUTRA modeling to outline freshwater–seawater mixing zones and calculate dispersivity. In a 100 × 50 × 10 cm flume with inclined layers and mixed barriers, Sharma and Chakma (2024) achieved a 40–48% reduction in the SWI front and delayed intrusion by up to 86.7%. More recently, Al-Hinai et al. (2025) used layered sandbox ASR trials to demonstrate that strategic positioning of injection and abstraction wells maintains higher withdrawal rates without degrading water quality and produces intricate, bubble-like mixing patterns.

Despite extensive prior work, no study has paired a full three-dimensional bench-scale flume with comprehensive geochemical investigation modeling for different pumping scenarios to assess the impact of various groundwater extraction scenarios in coastal areas on the extent of seawater intrusion and the resulting degradation in freshwater quality. Our research fills this gap by using real aquifer materials from Eastern Saudi Arabia and natural seawater and groundwater in controlled 3D flume experiments, collecting time-series samples under contrasting flows, and applying inverse PHREEQC modeling. This integrated approach allows us to simulate SWI under realistic conditions, quantify shifts in major ions and trace elements, and unravel the specific mixing, mineral-dissolution, and ion-exchange reactions driving groundwater chemistry changes, thereby delivering a predictive tool for managing seawater intrusion in arid coastal aquifers.

Methodology

The research methodology involves an integrated approach of lab-scale experiment simulation and inverse geochemical modeling to study the geochemical evolution of coastal groundwater in sandy aquifers impacted by seawater intrusion. The study's experimental layout and components were designed to duplicate shallow sandy coastal aquifers in urban arid coastal regions. Therefore, aquifer material and real groundwater from shallow aquifers in the eastern region of Saudi Arabia have been utilized in the seawater intrusion experiments. Moreover, seawater samples collected from the Arabian Gulf have been used in experiments to replicate the real-world situation. The groundwater and seawater chemical composition is shown in Table 1. The aquifer material was collected from a shallow sandy aquifer in eastern Saudi Arabia, with mineral composition illustrated in Table 2. The aquifer material is quartz-dominated with a significant amount of feldspar minerals and traces of calcite and heavy minerals.

Table 1—Table showing the chemical composition of real groundwater from the local aquifer in eastern Saudi Arabia and seawater from the Arabian Gulf used in the seawater intrusion experiments.

Parameter	Groundwater	Seawater
pH	7.33	7.66
EC ($\mu\text{S}/\text{cm}$)	6 463	68 480
TDS (mg/L)	4 136	43 827
Ca^{2+} (mg/L)	384.0	811.0
Mg^{2+} (mg/L)	94.9	1 838.5
Na^+ (mg/L)	751.6	12680
K^+ (mg/L)	44.2	652.3
Cl^- (mg/L)	1 459.5	24 664.5
HCO_3^- (mg/L)	59.3	185
Br^- (mg/L)	5.22	76.01
SO_4^{2-} (mg/L)	650.7	4430
B+ (mg/L)	0.65	4.4

Table 2—Mineral composition of the aquifer material collected from the local sandy aquifer in the eastern region of Saudi Arabia and used in the current seawater intrusion experiments.

Mineral	Composition %
Quartz	79.3
Halite	0.0
Sylvine	0.4
Aragonite	0.4
Rutile	1.2
Orthoclase	3.7
Calcite	0.6
Oligoclase	2.2
Anorthite	1.8
Muscovite	0.9
Microcline	3.2
Dolomite	0.3
Albite	2.1
Sanidine	3.2

Mineral	Composition %
Cordierite	0.6

In the first stage, two controlled seawater intrusion experiments were carried out using a bench-scale sandbox to simulate the seawater intrusion in coastal sandy aquifers (Fig. 1). A transparent, custom-built acrylic flume was used to simulate seawater intrusion (SWI) in a coastal aquifer system. The setup consists of three main compartments: a seawater reservoir (right), a sand-filled aquifer chamber (center), and a freshwater reservoir (left). Both the seawater (SW) and groundwater (GW) tanks have dimensions of 20 cm in length, 20 width, and 50 cm in height and serve as hydraulic boundaries for inducing horizontal flow. The central part of the sandbox, with a length of 80 cm, a width of 20 cm, and a height of 50 cm, was filled with aquifer material. Different pumping scenarios (high (12 mL/minute) and low (6 mL/minute)) associated with no recharge conditions have been experimented with to reveal the evolution of groundwater in coastal sandy aquifers in arid regions. Water samples have been collected regularly to delineate the change in water quality due to seawater intrusion. The water samples were analyzed for physicochemical properties, ion composition, and trace and toxic elements.

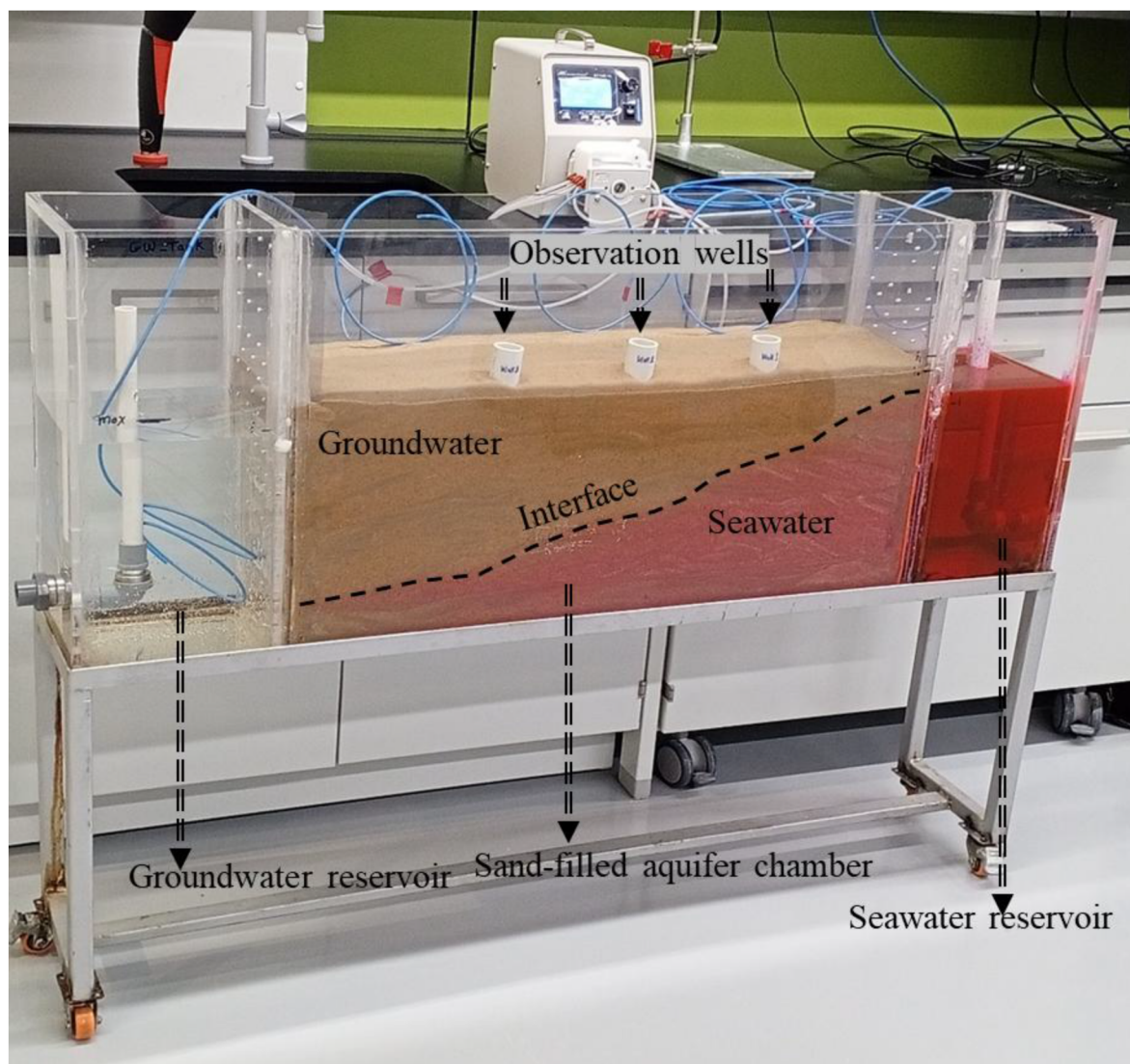


Figure 1—Laboratory-Scale Physical Model Developed for Simulating Seawater Intrusion in a Coastal Aquifer.

The sandbox experiments were conducted using 65 kg of aquifer sand, packed into a transparent acrylic flume measuring 80 cm × 20 cm × 50 cm. Groundwater was collected on-site in eastern Saudi Arabia, and seawater was drawn from Corniche Al Khobar; each run used 25 L of groundwater and 40 L of seawater to establish the hydraulic tanks. Two experiments spanned pumping rates of 6 and 12 mL/min, with each run lasting up to four days and samples drawn at EC change intervals. All valves and pumps were calibrated to maintain steady flow and avoid turbulence, and photographs of each setup were archived to document the precise configuration for replicability.

Water samples were analyzed for major anions (SO_4^{2-} , Cl^- , Br^- , NO_2^-) by high-performance ion chromatography following EPA Method 9056A and for cations and trace metals (B , Mg^{2+} , K^+ , Na^+ , Ca^{2+}) by ICP-MS in accordance with US EPA Method 6020. Electrical conductivity and temperature were monitored in real time using a HERON Instruments Conductivity Plus probe, and pH was measured with an EUTECH PCD650 meter, both calibrated before each run. Total inorganic carbon (TIC) was determined via a TOC analyzer and converted to carbonate speciation using the modified Henderson–Hasselbalch equation to infer HCO_3^- and CO_3^{2-} concentrations based on measured pH. These methods together ensured high-resolution, time-series quantification of hydrochemical evolution under contrasting pumping regimes.

The experimental data have been utilized to develop inverse geochemical models using PHREEQC software version 3 (<https://www.usgs.gov/software/phreeqc-version-3>) to delineate the geochemical evolution of coastal groundwater in sandy aquifers impacted by seawater intrusion.

Results and Discussion

Geochemical evolution of coastal groundwater during a high-pumping scenario.

The high pumping scenario of groundwater from the coastal sandy aquifer and the impact of seawater intrusion on the geochemical evolution have been experimented by implementing a pumping rate of 12 mL/minute in a lab-scale sandbox physical model (Fig. 1). The result of high pumping experiment of a feldspar-rich sand layer showed significant seawater intrusion, causing permanent changes in groundwater quality. The results indicate a sharp increase in electrical conductivity (EC) of groundwater from 5752 $\mu\text{S}/\text{cm}$ in the control sample to more than 44,000 $\mu\text{S}/\text{cm}$ in the wells located along the flow path, along with parallel increases in total dissolved solids (TDS) and major ions, with sodium exceeding 9800 mg/L and chloride rising to over 15,300 mg/L in the most affected samples. Such an increase in the ion composition of SWI-impacted groundwater reflects the role of the seawater-groundwater mixing process. The slightly elevated concentration of Na^+ cation with a reduction in divalent cations (Ca^{2+} and Mg^{2+}), especially Ca^{2+} cation may indicate the impact of the reverse cation exchange mechanism on the chemistry of coastal groundwater impacted by seawater intrusion (Fig. 2). In high pumping scenarios, the ion exchange happened after the seawater intrusion magnitude reached 36% with EC of 25000 $\mu\text{S}/\text{cm}$.

While we did not directly measure bulk CEC via an extraction method, we quantified ion-exchange behavior through high-resolution water chemistry coupled with mineralogical analysis of the aquifer material. Powder X-ray diffraction of our sand sample as shown in Table 2 confirmed the presence of feldspar phases and minor clay minerals—phases known to exhibit CEC far above that of "clean" quartz sand. During the 12 mL/min runs, we observed a clear threshold at ~36 % seawater mixing where Ca^{2+} concentrations rapidly declined while Na^+ rose (Figs. 3–4). The XRD-inferred exchangeable mineral abundance and the distinct shifts in divalent/monovalent cations demonstrate active cation exchange well beyond what quartz alone could support. After this threshold value, the Ca^{2+} concentration decreased due to the reverse cation exchange process, which continued to the end of the experiment Fig. 3.

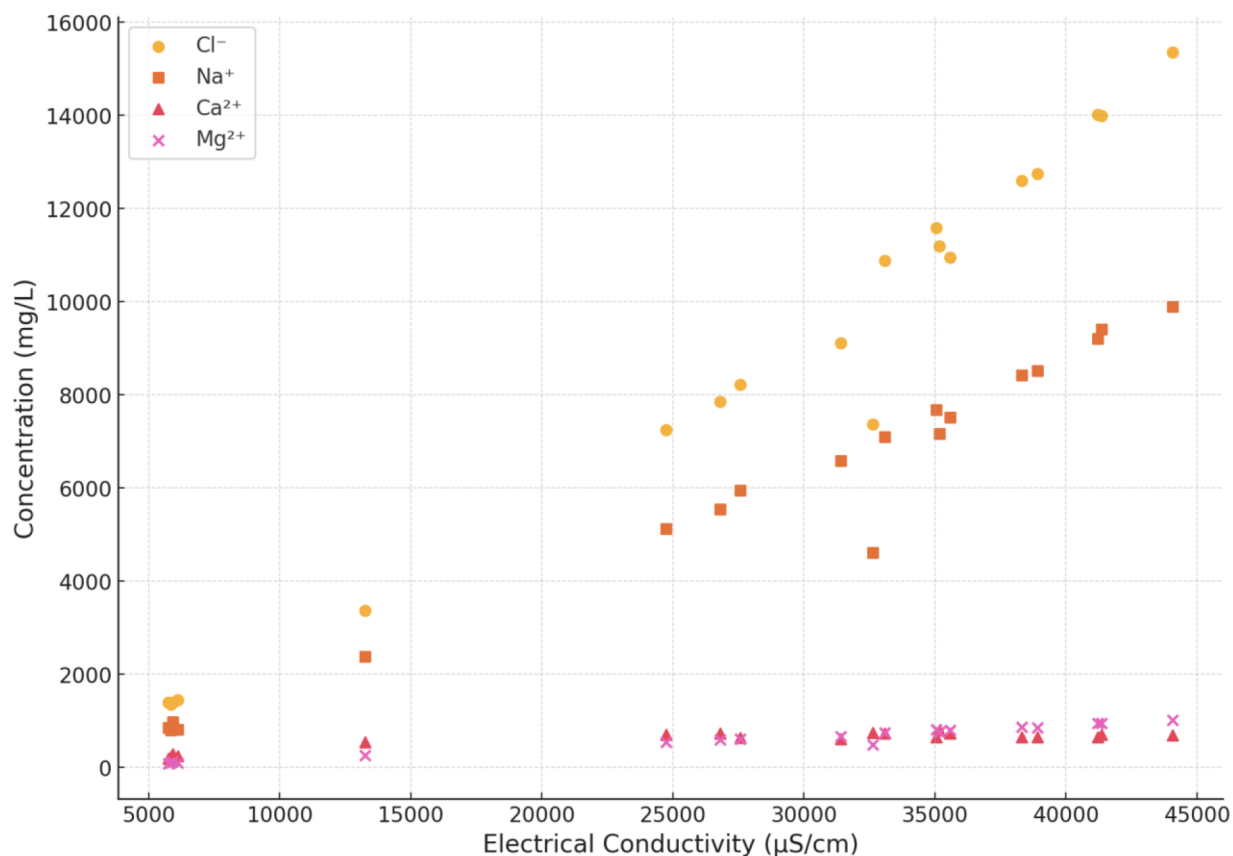


Figure 2—Water ion changes during seawater intrusion experiment with high pumping scenario.

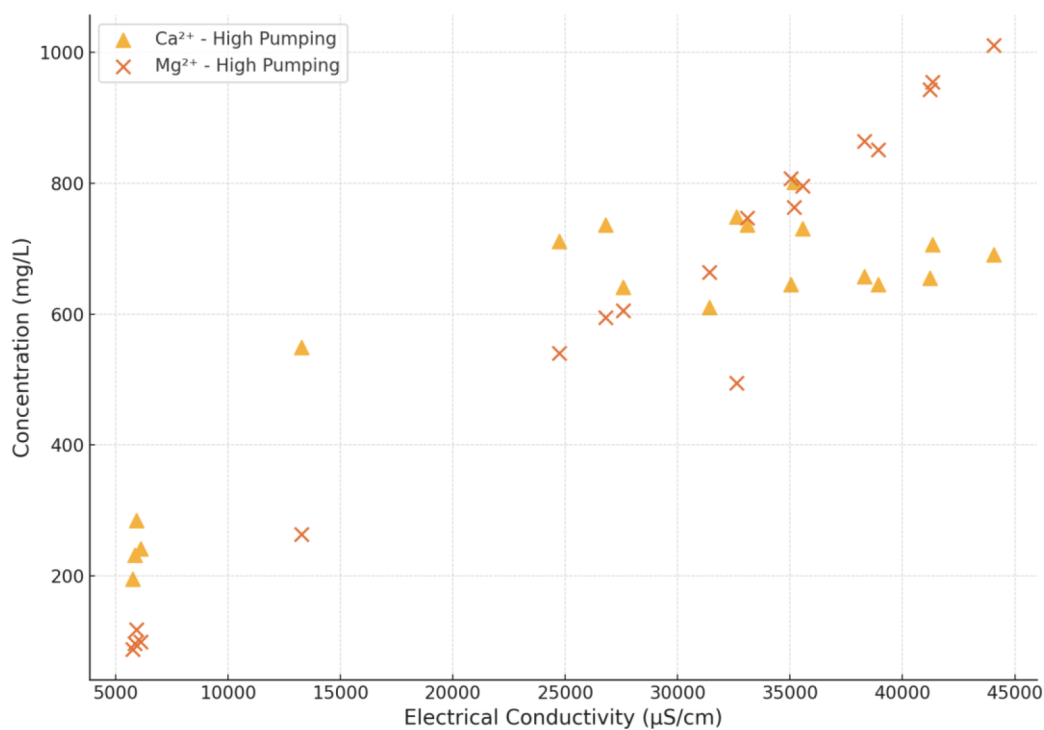


Figure 3—Ca and Mg concentration in water samples impacted by different degrees of seawater intrusion with high pumping scenario

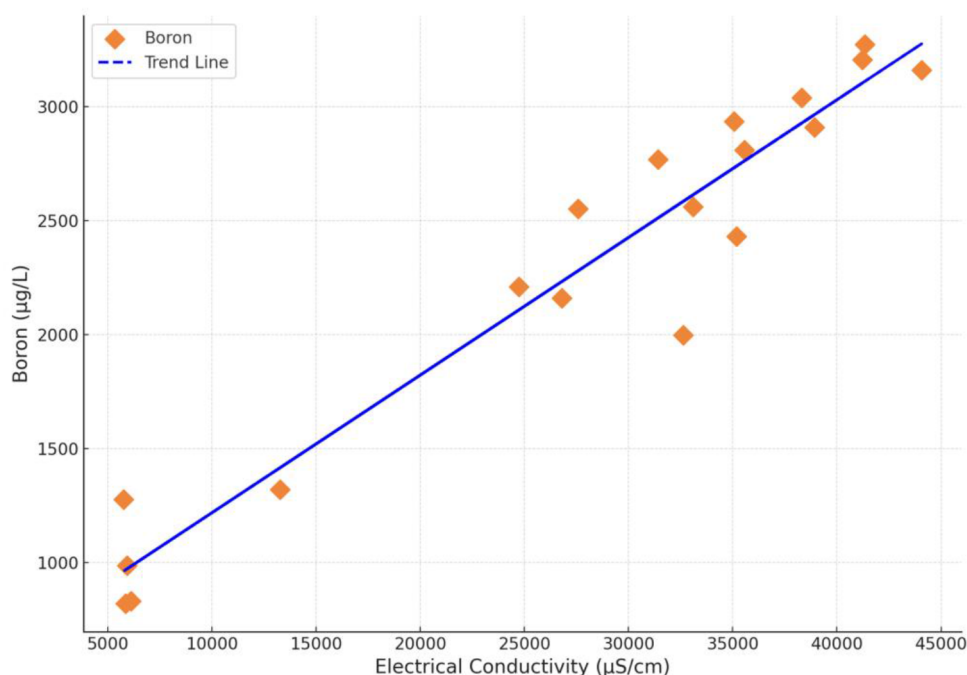


Figure 4—Boron concentration in water samples impacted by different degrees of seawater intrusion during high pumping scenario.

Among the trace and toxic elements, the elevated concentration of boron in the Arabian Gulf and its adverse impact on human health motivated us to investigate its mobility and behavior through the feldspar-rich sandy material seawater intrusion experiments. Boron concentrations showed increased from 650 µg/L in the groundwater control sample to around 3200 µg/L in the affected wells, reflecting the trace element mobility in feldspar-rich sand layers with poor boron absorption capacity Fig. 4. Increase boron concentration in groundwater samples impacted by seawater intrusion with magnitude of 6%, which means with increase groundwater salinity by 1000 µS/cm, the boron level increased by 60 µg/L. Such observations reflect the degradation rate of groundwater quality due to elevated boron concentrations that may have adverse effects on the human health of coastal communities.

Geochemical evolution of coastal groundwater during a low-pumping scenario.

A low pumping rate of 6 mL/Minute was used in the second experiment of the seawater intrusion physical model simulation. This experiment was conducted to reveal the change in groundwater chemistry and quality and to identify the geochemical process-controlled groundwater evolution. The experiment result showed a gradual increase in electrical conductivity (EC) of groundwater from 57 52 µS/cm in the control sample to more than 54000 µS/cm. The EC increase is associated with proportional increases in total dissolved solids (TDS) and major ions, such as sodium (Na⁺) and chloride (Cl⁻), with sodium exceeding 11000 mg/L and chloride rising to over 18000 mg/L Fig. 5. The groundwater samples in well close to seawater reservoir reach EC value of seawater, however, the well close to groundwater reservoir showed water with EC value less than the seawater. As shown in Fig. 6, the Ca²⁺ concentration increased with the increase of the EC value due to seawater intrusion. However, the increase of Ca²⁺ concentration was not proportional to the increase of EC value, indicating the impact of the reverse cation exchange process during the seawater intrusion experiment of the low pumping scenario. In this experiment, the cation exchange process occurred gradually and continued to the end of the experiment.

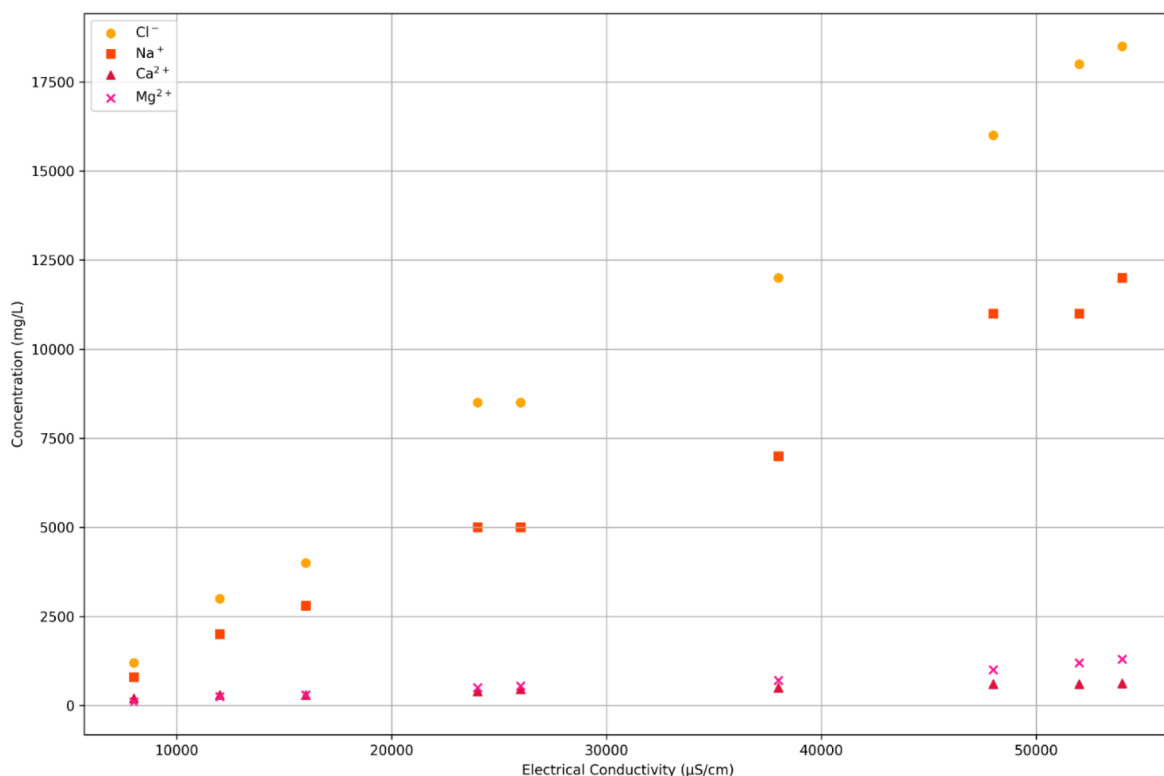


Figure 5—Water ion changes during the seawater intrusion experiment with low pumping scenario.

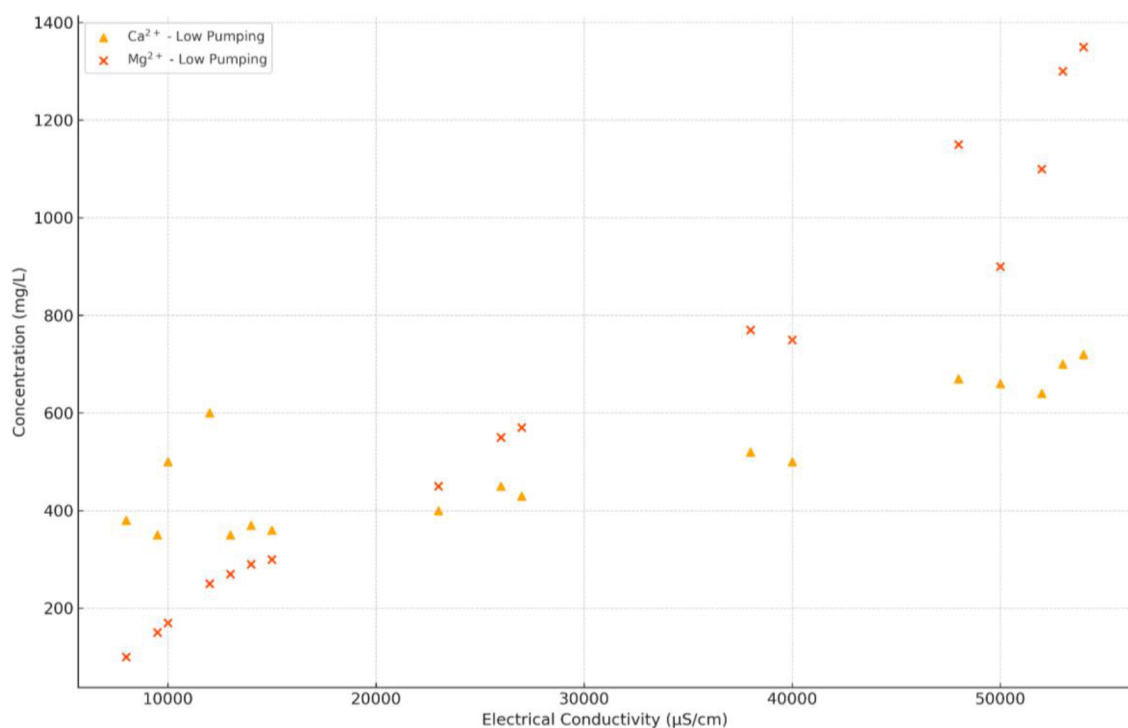


Figure 6—Ca and Mg concentration in water samples impacted by different degrees of seawater intrusion with low pumping scenario.

As illustrated in Fig. 7, boron concentration has changed during the low pumping scenario and displays a gradual increase with a rate of 4.4%. This means that by increasing the groundwater salinity by 1000 $\mu\text{S}/\text{cm}$, the boron concentration increased by 44 $\mu\text{g}/\text{L}$. The magnitude of boron concentration change during the

low pumping scenario is lower than boron concentration changes in the high pumping scenario by $12 \mu\text{g/L}$ for each $1000 \mu\text{S/cm}$ change in groundwater salinity. Therefore, the higher groundwater extraction from the coastal region is most likely associated with an increase in boron concentration in the SWI-impacted aquifer and may lead to a decline the groundwater quality and its suitability for human use.

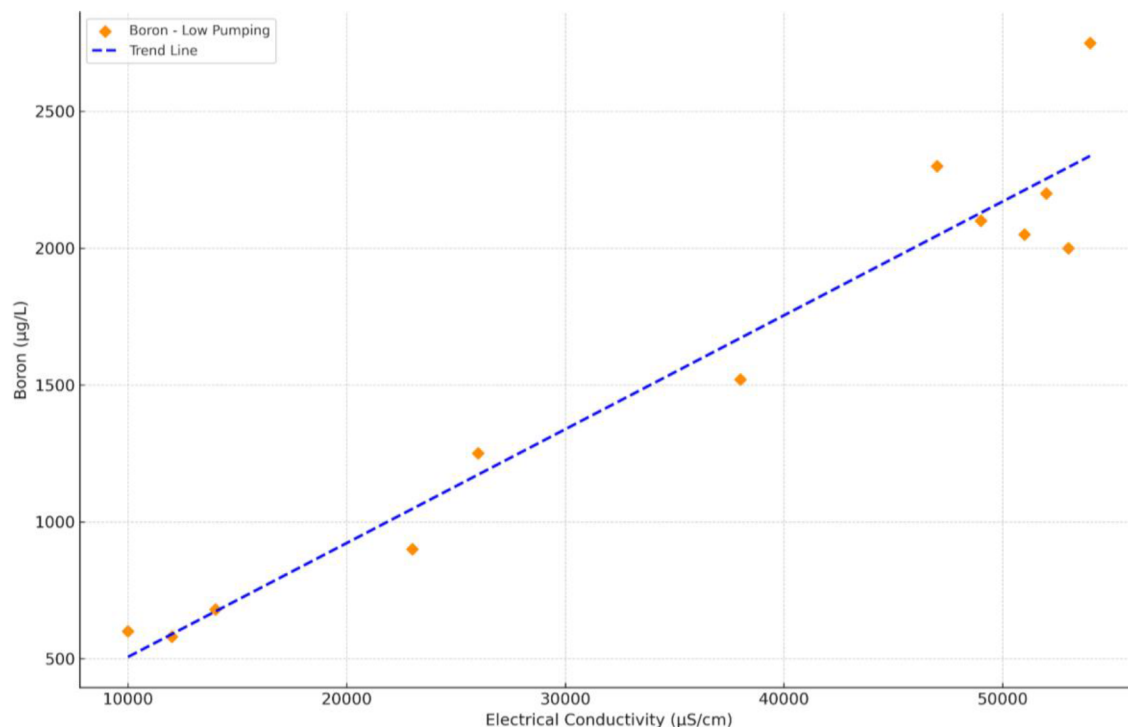


Figure 7—Boron concentration in water samples impacted by different degrees of seawater intrusion during the low pumping scenario.

Inverse geochemical modeling.

An inverse geochemical model has been developed using PHREEQC (Parkhurst & Appelo, 2013) to reconstruct the hydrogeochemical evolution of groundwater subject to seawater intrusion in the current study. Moreover, the model aided us in quantifying the mixing rate between fresh groundwater and seawater, as well as the geochemical processes to reproduce the observed composition of a representative salinized groundwater sample from low pumping experiments. Three end-member solutions, including a groundwater control sample, a seawater control sample, and a SWI-impacted groundwater sample, were used to develop the inverse geochemical model. The model was constructed using the chemical composition of these three endmembers with parameters including major ions Ca^{2+} , Mg^{2+} , Na^+ , K^+ , Cl^- , SO_4^{2-} , Br^- , and alkalinity as HCO_3^- , water density, and water pH. The geochemical simulation was conducted using the phreeqc.dat database, which is capable of all calculations of solution concentration. The solid phase assemblage was taken directly from our powder X-ray diffraction results (Table 2) and includes quartz (inert), feldspar minerals (orthoclase, oligoclase, and anorthite totaling $\sim 15\%$, carbonate phases (calcite, dolomite, and aragonite), and accessory evaporites (halite and gypsum). These minerals were allowed to dissolve or precipitate as needed to satisfy mass-balance constraints in each batch simulation.

Because we performed "in-glass" (batch) inversions rather than one-dimensional reactive transport runs, there is no advective dispersion term—mixing is imposed by a linear combination of endmembers in the reaction block. We did not directly measure bulk cation exchange capacity (CEC) by conventional extraction, but our mineralogy implies the presence of exchange-active surfaces: feldspars and minor clay

phases support far higher CEC than pure quartz sand. Cation-exchange behavior begins at ~36% seawater mixing ($EC \approx 25,000 \mu S/cm$) and is constrained.

In each PHREEQC run, solution compositions (pH, T, major ions, and alkalinity) came from our EC-driven sampling schedule, and all calculations employed the USGS phreeqc.dat database. We prescribed equilibrium with quartz (to buffer silica activity), halite, gypsum, and calcite, and allowed feldspar dissolution implicitly through the inverse-model fitting of Na^+ , K^+ , Ca^{2+} , and Mg^{2+} trends. A small balance tolerance (1×10^{-6}) and uncertainty factor (0.1) ensured that the optimized combination of dissolution, precipitation, mixing, and exchange reactions reproduced the observed composition with minimal residuals. This schematization captures the dominant geochemical controls—simple mixing, mineral reactions, and surface exchange—without invoking full transport coupling and provides a validated framework for predicting groundwater chemistry under varying SWI scenarios.

The simulation has successfully reached the optimal solution that best fits the measured composition of the experimental water sample, considering the physical mixing and possible reactions. The results showed that the observed composition of the experimental sample can be regenerated by mixing seawater with groundwater endmembers with ratios around 59% and 41%, respectively. Minor calcite and gypsum dissolution were observed with no significant impact on groundwater chemistry. Therefore, the mixing process is dominant in seawater intrusion experiments of sandy aquifers, and it accounts for changes in groundwater quality and chemistry. As illustrated in Fig. 8, the modeled concentrations of SWI-impacted water samples for all tested ions are matched with measured concentrations from an experiment with minor error. The minor discrepancy between the measured concentration and model concentration most probably reflects the impact of the reverse cation exchange process, which has been documented and illustrated in the experimental part.

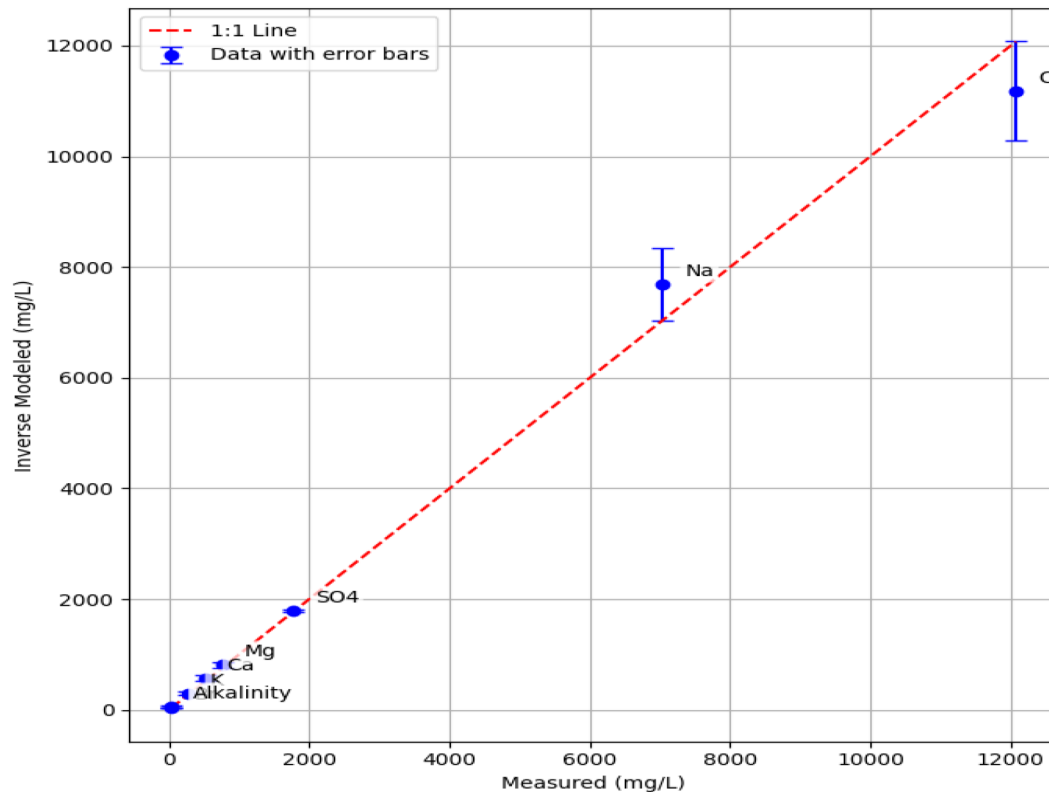


Figure 8—Measured versus modeled concentration of water samples impacted by seawater intrusion during the low pumping scenario.

The study results confirm that the principal process that controls groundwater chemistry and quality in coastal sandy aquifers impacted by seawater intrusion is the mixing process between seawater and groundwater. These findings are compatible with previous works investigating the geochemical mechanism of groundwater in coastal aquifers (Benaafi et al., 2023b; Y. Liu et al., 2017; Salem & Osman, 2017; Wang et al., 2022). Furthermore, the reverse cation exchange that has been observed during the experimental work of this study is considered the secondary process for groundwater chemistry change as documented in the literature by several researchers (Ayari et al., 2023; Benaafi et al., 2022; Moorthy et al., 2024; Morgan et al., 2013).

Conclusions

The current study implemented integrated physical and geochemical modeling of groundwater evolution in coastal sandy aquifers impacted by seawater intrusion in urban and arid regions. High and low pumping scenarios have been applied to reflect the real-world situation. The results indicate that the main process that controls the groundwater evolution in coastal aquifers is the mixing process. However, the progression of seawater into coastal aquifers during high pumping led to clear and sharp changes in water chemistry due to the reverse cation exchange process after the threshold value representing 36% of seawater-groundwater mixing. In contrast, during the low pumping scenario, the impact of the reverse cation exchange process was gradual with low magnitude compared with the high pumping scenario. Moreover, trace elements such as boron have shown high mobility during high-pumping scenarios compared with low-pumping conditions. This concluded that groundwater evolution during the high pumping scenario of the coastal aquifer led to degrading water quality and its suitability for irrigation and drinking due to a substantial increase in sodium contents and boron concentration. The inverse geochemical model was able to simulate the change in water chemistry and can be used as an effective predictive tool of water quality in coastal aquifers impacted by seawater intrusion. Such models will help the decision maker to develop effective plans to minimize the impact of seawater on coastal water resources and safeguard public health.

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